

How to compute the average inverse exit-time ...

From "Nonasymptotic properties of transport and mixing" by Boffetta, Celani, Cencini, Lacorata and Vulpiani (Chaos 10, 50-60, 2000):

The average value of the inverse exit-time ($1/\tau_n$) from shell n to shell $n+1$, over a large number of realizations, $M \gg 1$, is computed by considering that the probabilistic weight of the outcome $1/\tau_n^{(i)}$ is proportional to the duration of the outcome:

$$p(\tau_n^{(i)}) = \tau_n^{(i)} / \sum_{i=1}^M \tau_n^{(i)}$$

$$\left\langle \frac{1}{\tau_n} \right\rangle = \frac{\sum_{i=1}^M \tau_n^{(i)} (1/\tau_n^{(i)})}{\sum_{i=1}^M \tau_n^{(i)}} = \frac{M}{\sum_{i=1}^M \tau_n^{(i)}} = \frac{1}{\langle \tau_n \rangle}$$

The variance of the inverse exit-time is computed according to the same procedure

$$\left\langle \frac{1}{\tau_n^2} \right\rangle = \frac{\sum_{i=1}^M \tau_n^{(i)} \left(1/\tau_n^{(i)}\right)^2}{\sum_{i=1}^M \tau_n^{(i)}} = \frac{\sum_{i=1}^M \left(1/\tau_n^{(i)}\right)}{\sum_{i=1}^M \tau_n^{(i)}} = \frac{\langle 1/\tau_n \rangle}{\langle \tau_n \rangle}$$

$$\sigma^2 \left(\frac{1}{\tau_n} \right) = \left\langle \frac{1}{\tau_n^2} \right\rangle - \frac{1}{\langle \tau_n \rangle^2} \quad \text{variance of the single outcome}$$

$$\hat{\sigma} \left(\left\langle \frac{1}{\tau_n} \right\rangle \right) = \sigma \left(\frac{1}{\tau_n} \right) / \sqrt{M} \quad \text{standard deviation of the mean outcome}$$

$$\bar{\sigma}(\delta) = \hat{\sigma} \left(\left\langle \frac{1}{\tau} \right\rangle \right) \ln(r) \quad \text{standard deviation of the FSLE at given } \delta, \langle \tau(\delta) \rangle \text{ and } r$$

$$\text{error bar on the FSLE: } \lambda(\delta) \pm \bar{\sigma}(\delta)$$